INFOSYS SE PSEUDOCODE VAULT • ARRAYS, STRINGS & DATA STRUCTURES

1D/2D Arrays, Matrices, Element Swapping, String Indexing • Total Solved Questions: 41

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Index Calculation

Q1. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 2, 9, 5]
display a[a[2]] + a[1]
```

(A) 7
(B) 9
(C) 11
(D) 13

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** a[2]=2, so a[a[2]]=a[2]=2; 2+7=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Transformation

Q2. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
for i = 0 to 3 step 1 do
a[i] = a[i] + i
end-for
display a[3] - a[1]
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (D) : 6

■ **Step-by-Step Logic:** After updates, a=[3,6,9,12]; 12−6=6.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Maximum

Q3. What will be the output of the following pseudocode?

```
Set Integer a = [8, 3, 15, 6, 11]
Set Integer m = a[0]
for i = 1 to 4 step 1 do
if (a[i] > m)
m = a[i]
end-if
end-for
display m % 4
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (C) : 3

■ **Step-by-Step Logic:** Maximum is 15; 15%4=3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Sum

Q4. What will be the output of the following pseudocode?

```
Set Integer a = [2, 5, 8, 11]
Set Integer s = 0
for i = 0 to 2 step 1 do
s = s + a[i] + a[i+1]
end-for
display s
```

(A) 36
(B) 39
(C) 42
(D) 45

Correct Answer: Option (B) : 39

■ **Step-by-Step Logic:** Pairs: 7+13+19=39.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Conditional Update

Q5. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 10, 13]
for i = 0 to 3 step 1 do
if (a[i] % 2 == 0)
a[i] = a[i] / 2
else
a[i] = a[i] + 1
end-if
end-for
display a[0] + a[1] + a[2] + a[3]
```

(A) 25
(B) 27
(C) 29
(D) 31

Correct Answer: Option (C) : 29

■ **Step-by-Step Logic:** Array becomes [2,8,5,14]; sum=29.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Reverse

Q6. What will be the output of the following pseudocode?

```
Set Integer a = [3, 6, 9, 12, 15]
display a[4] - a[0] + a[3] - a[1]
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 15−3+12−6=18. Infosys Medium Pseudocode Practice • Page 39

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Count

Q7. What will be the output of the following pseudocode?

```
Set Integer a = [5, 10, 15, 20, 25]
Set Integer c = 0
for i = 0 to 4 step 1 do
if (a[i] % 10 == 5)
c = c + 1
end-if
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** 5,15,25 satisfy the condition: 3.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Running Sum

Q8. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8]
Set Integer s = a[0]
for i = 1 to 3 step 1 do
a[i] = a[i] + s
s = a[i]
end-for
display a[3]
```

(A) 16
(B) 20
(C) 24
(D) 30

Correct Answer: Option (B) : 20

■ **Step-by-Step Logic:** Running values become 2,6,12,20; a[3]=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Nested Index

Q9. What will be the output of the following pseudocode?

```
Set Integer a = [1, 3, 0, 2]
display a[a[3]] + a[a[1]]
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** a[3]=2, so a[2]=0; a[1]=3, so a[3]=2; total=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Difference

Q10. What will be the output of the following pseudocode?

```
Set Integer a = [12, 5, 9, 4]
Set Integer m = a[0]
for i = 1 to 3 step 1 do
if (a[i] < m)
m = a[i]
end-if
end-for
display a[0] - m
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** Minimum is 4; 12−4=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Shift

Q11. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
display (a[3] >> 1) + (a[0] << 1)
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** 9>>1=4; 3<<1=6; total=10.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array XOR

Q12. What will be the output of the following pseudocode?

```
Set Integer a = [6, 10, 12, 3]
display (a[0] ^ a[1]) + a[3]
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** 6^10=12; 12+3=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Pair Comparison

Q13. What will be the output of the following pseudocode?

```
Set Integer a = [4, 9, 6, 11]
Set Integer c = 0
for i = 0 to 2 step 1 do
if (a[i] < a[i+1])
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** The true comparisons are 4<9 and 6<11, so count=2.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Sum with Step

Q14. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8, 10, 12]
Set Integer s = 0
for i = 0 to 5 step 2 do
s = s + a[i]
end-for
display s
```

(A) 12
(B) 16
(C) 18
(D) 20

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** Indices 0,2,4 give 2+6+10=18. Infosys Medium Pseudocode Practice • Page 40

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Array Rotation Logic

Q15. What will be the output of the following pseudocode?

```
Set Integer a = [10, 20, 30, 40]
Set Integer t = a[3]
for i = 3 to 1 step -1 do
a[i] = a[i-1]
end-for
a[0] = t
display a[1] + a[3]
```

(A) 30
(B) 40
(C) 50
(D) 60

Correct Answer: Option (B) : 40

■ **Step-by-Step Logic:** After right rotation: [40,10,20,30]; a[1]+a[3]=40.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length

Q16. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer n = stringLength(s)
display (n << 1) - 3
```

(A) 13
(B) 15
(C) 17
(D) 19

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** Length=9; 9<<1=18; 18−3=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Indexing

Q17. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[1] + s[5]
```

(A) NS
(B) NY
(C) FY
(D) SY

Correct Answer: Option (B) : NY

■ **Step-by-Step Logic:** s[1]=N and s[5]=Y, so the result is NY.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Condition

Q18. What will be the output of the following pseudocode?

```
Set String s = "COMPUTER"
if (stringLength(s) % 2 == 0)
display stringLength(s) / 2
else
display stringLength(s) + 2
end-if
```

(A) 4
(B) 5
(C) 6
(D) 10

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** COMPUTER has 8 characters; 8/2=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String + Loop

Q19. What will be the output of the following pseudocode?

```
Set String s = "CODE"
Set Integer total = 0
for i = 0 to stringLength(s)-1 step 1 do
total = total + i
end-for
display total
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (C) : 6

■ **Step-by-Step Logic:** Indices 0+1+2+3=6.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Index + Shift

Q20. What will be the output of the following pseudocode?

```
Set String s = "TESTING"
Set Integer n = s[1] == 'E'
display stringLength(s) << n
```

(A) 7
(B) 14
(C) 21
(D) 28

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** The comparison is true, represented as 1; length 7<<1=14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Modulo

Q21. What will be the output of the following pseudocode?

```
Set String s = "ALGORITHM"
display stringLength(s) % 4
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** ALGORITHM has 9 characters; 9%4=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Reverse Access

Q22. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[stringLength(s)-1]
```

(A) I
(B) S
(C) Y
(D) Error

Correct Answer: Option (B) : S

■ **Step-by-Step Logic:** Last index is 6; s[6] = S. Infosys Medium Pseudocode Practice • Page 41

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Character Count

Q23. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer c = 0
for i = 0 to stringLength(s)-1 step 1 do
if (s[i] == 'E')
c = c + 1
end-if
end-for
display c
```

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: Option (B) : 2

■ **Step-by-Step Logic:** PLACEMENT contains E twice.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Length + Right Shift

Q24. What will be the output of the following pseudocode?

```
Set String s = "INTERVIEW"
display stringLength(s) >> 1
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** INTERVIEW has 9 characters; 9>>1=4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: String Conditional

Q25. What will be the output of the following pseudocode?

```
Set String s = "NQT2026"
if (stringLength(s) > 6)
display s[0]
else
display s[1]
end-if
```

(A) N
(B) Q
(C) T
(D) 2

Correct Answer: Option (A) : N

■ **Step-by-Step Logic:** Length is 7, so the first branch displays s[0]=N.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Array Formula

Q26. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b, Integer c)
return (a + b) ^ c
End function
display f(5, 3, 6)
```

(A) 4
(B) 8
(C) 12
(D) 14

Correct Answer: Option (D) : 14

■ **Step-by-Step Logic:** 5+3=8; 8 XOR 6 = 14.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Function

Q27. What will be the output of the following pseudocode?

```
Integer f(Integer x)
return (x >> 1) + 3
End function
Set Integer a = [8, 12, 16]
display f(a[2])
```

(A) 9
(B) 10
(C) 11
(D) 12

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** a[2]=16; 16>>1=8; 8+3=11.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Array + Right Shift

Q28. What will be the output of the following pseudocode?

```
Set Integer a = [18, 25, 32, 41]
display (a[3] >> 2) + (a[0] % 5)
```

(A) 11
(B) 12
(C) 13
(D) 14

Correct Answer: Option (C) : 13

■ **Step-by-Step Logic:** 41>>2=10; 18%5=3; total=13. Infosys Medium Pseudocode Practice • Page 44

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + Array Update

Q29. What will be the output of the following pseudocode?

```
Set Integer a = [1, 2, 3, 4, 5]
for i = 1 to 4 step 1 do
a[i] = a[i] + a[i-1]
end-for
display a[4]
```

(A) 10
(B) 12
(C) 15
(D) 16

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Running cumulative values become [1,3,6,10,15].

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q30. What will be the output of the following pseudocode?

```
[Arrays]
What is printed?
Integer
A[4] = {3, 5, 7, 9}
Integer
i, p
Set
p = 1
for
i = 0 to 3 do
p
= p * A[i]
end
for
Print
p
```

(A) 24
(B) 315
(C) 945
(D) 105

Correct Answer: Option (C) : 945

■ **Step-by-Step Logic:** Product 3×5×7×9 = 945.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q31. What will be the output of the following pseudocode?

```
1.     Which of the following is a valid declaration of a
char?
```

(A) char ch = '\utea';
(B) char ca = 'tea';
(C) char cr = \u0223;
(D) char cc = '\itea';

Correct Answer: Option (A) : char ch = '\utea';

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : char ch = '\utea';.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q32. 1. What will be the output of this program fragment?

```
int i = 263;
      putchar(i);
```

(A) prints 263
(B) prints the ASCII equivalent of 263
(C) rings the bell
(D) prints garbage

Correct Answer: Option (C) : rings the bell

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (C) : rings the bell.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q33. What will be the output of the following pseudocode?

```
1.     Let A be a square matrix of size nxn. Consider the
following program and state the output.
C=100
for i= 1 to n do
for j = 1 to n do
{
Temp = A[i][j] + C
A[i][j] = A [j][i]
A[j][i]= Temp – C
}for i = 1 to n do
for j = 1 to n do

Output (A[i][j]);
```

(A) a) The matrix A itself
(B) a) Transpose of matrix A
(C) a) Adding 100 to the upper diagonal elements and subtracting 100 from diagonal elements of A
(D) a) None of the above

Correct Answer: Option (A) : a) The matrix A itself

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : a) The matrix A itself.

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q34. What will be the output of the following pseudocode?

```
1. Integer a = 5, b = 9 2. a = a ^ b 3. b = a ^ b 4. a = a
^ b 5. if (a == 9 && b == 5) 6. Print "Swapped" 7. else
8. Print "Not swapped"
```

(A) Line 2
(B) Line 3
(C) Line 4
(D) No error

Correct Answer: Option (D) : No error

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (D) : No error.

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q35. What will be the output of the following pseudocode?

```
What is the final content of arr? (indexing starts at 0;
updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
        arr[i] = arr[i] + arr[arr[i] % 4]
end for
```

(A) [8, 2, 5, 7]
(B) [8, 2, 5, 6]
(C) [8, 5, 5, 7]
(D) [5, 2, 5, 7]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q36. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
arr = [1, 2, 3, 4, 5]
temp = arr[0]
arr[0] = arr[4]
arr[4] = arr[2]
arr[2] = arr[1]
arr[1] = temp
```

(A) [5, 1, 2, 4, 3]
(B) [5, 1, 3, 4, 2]
(C) [5, 2, 1, 4, 3]
(D) [5, 1, 2, 4, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q37. What will be the output of the following pseudocode?

```
What is the final content of arr? (0-indexed; each
statement uses the array's state at that point)
```

(A) [60, 40, 30, 20, 50, 10]
(B) [60, 20, 30, 40, 50, 10]
(C) [10, 40, 30, 20, 50, 60]
(D) [60, 40, 50, 20, 30, 10]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q38. How many swaps are performed in total?

```
arr = [5, 2, 8, 1, 9, 3]
swaps = 0
for i = 0 to 4
        for j = 0 to 4 - i
                if (arr[j] > arr[j+1])
                        temp = arr[j]
                        arr[j] = arr[j+1]
                        arr[j+1] = temp
                        swaps = swaps + 1
                end if
        end for
end for
print(swaps)
```

(A) 7
(B) 8
(C) 6
(D) 9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q39. What will be the output of the following pseudocode?

```
This algorithm moves all non-zero elements to the front,
preserving order, and fills the rest with zeroes.
What is the final arr?
arr = [3, 0, 4, 0, 5, 0, 2]
pos = 0
for i = 0 to 6
        if (arr[i] != 0)
                arr[pos] = arr[i]
                pos = pos + 1
        end if
end for
while (pos <= 6)
        arr[pos] = 0
        pos = pos + 1
end while
print(arr)
```

(A) [3, 4, 5, 2, 0, 0, 0]
(B) [3, 0, 4, 0, 5, 0, 2]
(C) [3, 4, 5, 2, 0, 0, 2]
(D) [0, 0, 0, 3, 4, 5, 2]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q40. This is a binary search for target = 50 in a sorted array. How many comparisons are made before

```
the loop ends (the target is not present)?
arr = [2, 5, 8, 12, 16, 23, 38, 45, 56, 72, 91]
low = 0; high = 10; target = 50; comparisons = 0
while (low <= high)
        mid = (low + high) / 2
        comparisons = comparisons + 1
        if (arr[mid] == target)
                break
        else if (arr[mid] < target)
                low = mid + 1
        else
                high = mid - 1
        end if
end while
print(comparisons)
```

(A) 4
(B) 3
(C) 5
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q41. What will be the output of the following pseudocode?

```
This is Kadane's algorithm for maximum subarray sum. What
is printed?
arr = [-2, 3, -1, 5, -3, 4, -2, 6]
maxSoFar = arr[0]; maxEndingHere = arr[0]
for i = 1 to 7
        maxEndingHere = max(maxEndingHere + arr[i],
arr[i])
        maxSoFar = max(maxSoFar, maxEndingHere)
end for
print(maxSoFar)
```

(A) 12
(B) 10
(C) 9
(D) 15

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.